

(54) Title of the invention : ELECTRICAL LEAKAGE DETECTION SYSTEM

(51) International classification :G01R0031520000, G08B0021020000, G08B0007060000, B63B0079100000, G08B0025100000 (86) International Application No :NA Filing Date :NA (87) International Publication No : NA Filing Date :NA (61) Patent of Addition to Application Number :NA Filing Date :NA (62) Divisional to Application Number :NA Filing Date :NA	(71)Name of Applicant : 1)NITTE (DEEMED TO BE UNIVERSITY) Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA 575018 Dakshina Kannada - ----- Name of Applicant : NA Address of Applicant : NA (72)Name of Inventor : 1)SANDEEP KUMAR S Address of Applicant :DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING, NMAM INSTITUTE OF TECHNOLOGY, NITTE (DEEMED TO BE UNIVERSITY), NITTE - 574110, KARNATAKA, INDIA Udupi ----- 2)CHAITHRA S NAYAK Address of Applicant :DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING, NMAM INSTITUTE OF TECHNOLOGY, NITTE (DEEMED TO BE UNIVERSITY), NITTE - 574110, KARNATAKA, INDIA Udupi ----- 3)KHUSHI SHETTY Address of Applicant :DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING, NMAM INSTITUTE OF TECHNOLOGY, NITTE (DEEMED TO BE UNIVERSITY), NITTE - 574110, KARNATAKA, INDIA Udupi ----- 4)NIHARIKA SHETTY Address of Applicant :DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING, NMAM INSTITUTE OF TECHNOLOGY, NITTE (DEEMED TO BE UNIVERSITY), NITTE - 574110, KARNATAKA, INDIA Udupi ----- 5)ASHISH SINGH Address of Applicant :DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING, NMAM INSTITUTE OF TECHNOLOGY, NITTE (DEEMED TO BE UNIVERSITY), NITTE - 574110, KARNATAKA, INDIA Udupi ----- 6)RADHAKRISHNA Address of Applicant :DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING, NMAM INSTITUTE OF TECHNOLOGY, NITTE (DEEMED TO BE UNIVERSITY), NITTE - 574110, KARNATAKA, INDIA Udupi ----- 7)GURUMURTHY S C Address of Applicant :MANIPAL INSTITUTE OF TECHNOLOGY, MANIPAL ACADEMY OF HIGHER EDUCATION, MANIPAL - 576104 KARNATAKA Udupi ----- 8)RAMAPRASAD POOJARY Address of Applicant :SCHOOL OF ENGINEERING & IT, MANIPAL ACADEMY OF HIGHER EDUCATION, DUBAI CAMPUS, UAE -----
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(57) Abstract :

Disclosed herein is an electrical leakage detection system (100) for detecting current leakage in electric poles and providing hazard alerts comprises a plurality of sensors (102) to monitor electrical and environmental conditions, a display screen (110) to display real-time warnings, and a buzzer (108) to emit an audible alert, a communication network (114). A microcontroller (116) further comprises a data input module (118) to receive real-time data, a data preprocessing module (120) to filter and normalize raw data, a voltage monitoring module (122) to evaluate voltage data, a proximity calculation module (124) to compute the distance between the electric pole and nearby individuals and animals, an alert generation module (126) to activate the buzzer (108) and the display screen (110) and an output module (132) to communicate hazard information. The system also includes a user device (112) configured to receive remote alerts and real-time hazard notifications.

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